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PROVISIONAL/COMPLETE SPECIFICATION

1. TITLE OF THE PATENT

NECKBAND WITHOUT EARPIECES THAT SHOOTS DIRECTED SOUND

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in

which is to be performed

TITLE OF THE INVENTION: NECKBAND WITHOUT EARPIECES THAT SHOOTS DIRECTED SOUND

FIELD OF THE INVENTION

The present invention is generally related to the field of the Consumer Electronics/Wearable Technology.

BACKGROUND OF THE INVENTION

Traditional headphones and earbuds deliver sound directly into the ears but often cause discomfort, ear fatigue, and isolation from ambient sounds, making them unsuitable for situations requiring awareness, like commuting or outdoor activities.

Directed sound technology offers a solution by precisely delivering audio to a specific area without earpieces, but current systems are often bulky and not portable.

This invention introduces a neckband without earpieces that uses directed sound technology, providing a comfortable, immersive, and private listening experience while maintaining situational awareness. It offers a user-friendly, ergonomic alternative to traditional personal audio devices. Some of the reference known to me is as follows:

Prior Arts:

[01] Chinese patent application CN110506424B describes a neck speaker system and method designed to enhance audio delivery. The device, worn around the neck, utilizes directed sound technology to emit sound beams aimed at the user's ears. Unlike traditional speakers, this neck speaker allows users to enjoy a private and immersive listening experience without isolating them from their surroundings. The invention is particularly suited for scenarios requiring situational awareness, such as commuting or outdoor activities. The device's compact and ergonomic design ensures portability and comfort, making it an ideal audio solution for modern users.

[02] US patent application US10499183B2 describes a method and system for Bluetooth earbuds with enhanced connectivity and functionality, comprising a pair of earbuds, a Bluetooth communication module, an application server, and a user control interface. The system allows seamless pairing with multiple devices, prioritizes audio streams based on user preferences, and dynamically manages connection stability.
The Bluetooth communication module ensures secure transmission of data between the earbuds and connected devices using an access policy that restricts unauthorized devices from connecting. Additionally, the system can block connections or switch audio streams if conflicting requests are detected, ensuring a reliable and user-focused listening experience.

[03] US Patent Application US11800280B2 describes the method and system of a directional glass speaker, branded as Focusound, which utilizes advanced sound projection technology to deliver precise, focused audio output. The system comprises a transparent glass panel integrated with actuators, a sound processor, and a Bluetooth communication module. The primary purpose of the Focusound directional glass speaker is to project audio in a highly

targeted manner, creating a private listening zone while minimizing sound leakage. This innovation is particularly effective for applications such as public displays, smart home systems, and personal entertainment setups. The Focusound speaker also includes an interactive control interface embedded within the glass panel, allowing users to adjust settings intuitively. It communicates seamlessly with external[2] devices via Bluetooth, enhancing its usability and integration into modern connected environments.

State of the art suffers from the following **limitations**:

State-of-the-art neckbands without earpieces that shoot directed sound face several limitations.
[01] the technology relies heavily on precise sound direction, which can be challenging to achieve consistently across different environments and user movements.
[02] without the earphones in direct contact with the ears, sound leakage becomes a concern, as the sound may not remain confined to the intended area, leading to potential disturbances to others nearby.
[03] the lack of ear coverage can reduce the overall quality of the listening experience, particularly for bass frequencies.
[04] these devices might have limited compatibility with various audio sources or struggle to deliver optimal sound in noisy or outdoor environments.

OBJECTIVES OF THE INVENTION

The invention of a neckband without earpieces that shoots directed sound aims to deliver a personalized audio experience by focusing sound directly to the listener's ears, minimizing sound leakage and disturbance to others. It seeks to combine high-quality, directional sound with a comfortable, lightweight design, offering an alternative to traditional headphones and earphones. The objective is to ensure long battery life, clear audio reproduction, and ergonomic comfort while overcoming challenges such as interference, sound leakage, and limited range. Ultimately, the goal is to provide an immersive, discreet audio solution for users in various environments.

SUMMARY OF THE INVENTION

[01] The neckband without earpieces that shoots directed sound represents an innovative approach to personal audio technology. This device is designed to deliver sound directly to the listener's ears without the need for traditional earphones or headphones. The core idea is to utilize advanced sound projection technologies to aim audio precisely where it is needed—at the listener—while minimizing sound leakage to the surrounding environment. This ensures a private listening experience, ideal for shared or public spaces, allowing the wearer to enjoy audio content without disturbing others.
[02] One of the primary objectives of this invention is to provide a more comfortable alternative to conventional earphones or headphones. By eliminating the need for earpieces, the neckband offers a solution that avoids discomfort and pressure on the ears, making it suitable for extended use. Additionally, the neckband is designed to be lightweight and ergonomic, enhancing user comfort while offering a stylish, modern wearable audio device.
[03] The directed sound technology employed in the neckband aims to offer high-quality audio with minimal leakage, ensuring that only the wearer hears the sound clearly. This technology could utilize ultrasonic or beamforming techniques to create focused sound waves that travel directly toward the listener’s ears, regardless of the surrounding noise. As a result,

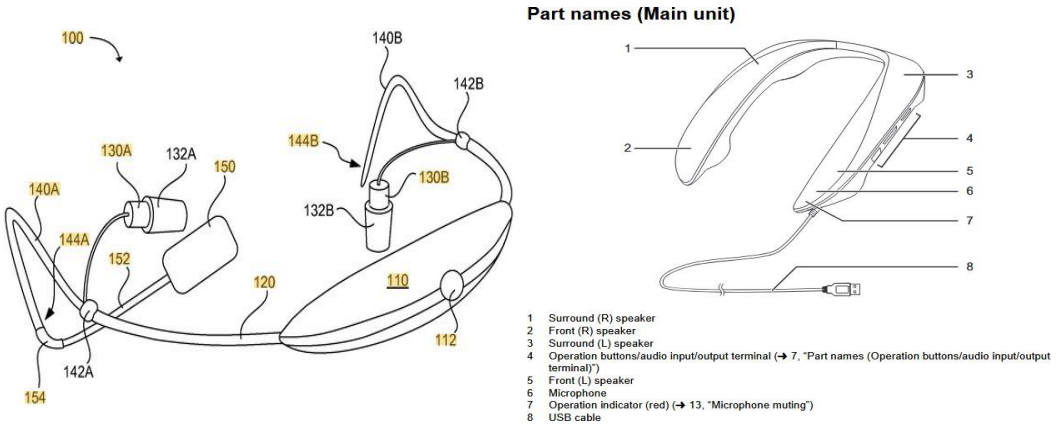
the device can provide a rich and immersive listening experience, even in noisy environments, without the need for earplugs or over-ear designs.

[04] Beyond audio quality, the invention also focuses on the user's safety and environmental awareness. Since the neckband doesn't block ambient sounds as traditional headphones do, users remain aware of their surroundings. This feature is particularly important for individuals who need to hear external sounds for safety reasons, such as pedestrians, cyclists, or drivers.

[05] The ability to listen to audio while staying attuned to environmental noises is a significant advantage.

Overall, this neckband represents a major step forward[3] in wearable audio technology, offering a combination of personalized sound, comfort, environmental awareness, and style.

DETAILED DESCRIPTION OF THE INVENTION



The neckband without earpieces that shoots directed sound is a novel wearable audio device designed to provide high-quality sound directly to the listener's ears without the need for traditional earphones or headphones. The device is primarily composed of a lightweight, ergonomic neckband that rests comfortably around the user's neck, allowing for a hands-free, wireless listening experience. The key feature of this neckband is its ability to use advanced sound projection technologies, such as ultrasonic or beamforming, to direct sound precisely toward the wearer's ears, ensuring privacy and preventing sound leakage into the surrounding environment.

[01] The neckband houses an array of miniature speakers or sound emitters strategically placed along its structure. These speakers are designed to emit sound in a highly focused manner, creating a beam of sound that travels in a specific direction—toward the listener's ears. The technology used ensures that the sound remains confined within this directional beam, offering clear audio to the wearer while minimizing interference from ambient noises. By utilizing directional sound techniques, the neckband avoids the need for ear contact, making it a suitable option for users who prefer not to have traditional earpieces in their ears or around their heads. In terms of sound quality, the device leverages high-fidelity audio technology, ensuring that the user experiences clear, rich, and immersive sound. The system is designed to reproduce a broad range of frequencies, from deep bass to high treble, while minimizing distortion and sound leakage. Because the sound is projected directly at the listener's ears, users can enjoy an enhanced audio experience without disturbing others in close proximity. The sound beam can be adjusted dynamically, with settings for volume, pitch, and even directional focus, allowing for a fully personalized listening experience.

[02] The neckband's design is another crucial element of this invention. It is engineered to be lightweight, ergonomic, and comfortable for prolonged use. The flexible neckband sits

around the user's neck, and the device may feature adjustable components, such as flexible arms or adjustable positions for the speakers, allowing the user to fine-tune the direction of the sound.

[03] The design ensures that the device remains secure during physical activity, such as walking, jogging, or cycling, without shifting or causing discomfort. Additionally, the neckband can be equipped with Bluetooth or other wireless connectivity options, making it compatible with smartphones, tablets, and other audio sources.

An important consideration for this invention is the balance between delivering high-quality sound and maintaining environmental awareness. Unlike traditional earphones or headphones, which can isolate the wearer from external sounds, the directed sound technology allows the user to remain aware of their surroundings. This is particularly beneficial for safety in outdoor environments where hearing traffic, pedestrians, or other potential hazards is essential. For instance, pedestrians, cyclists, or runners can enjoy their music while still being alert to the sounds of their environment, reducing the risk of accidents.

[04] The neckband is designed with user customization in mind. It can be equipped with controls to adjust the sound direction, volume, and even the type of audio playback. Some models might feature voice commands or touch-sensitive controls for ease of use while on the go. The device may also include sensors that detect the listener's head position or movements, automatically adjusting the sound direction for optimal listening as the user moves.

[05] The power system of the neckband is optimized for efficiency, incorporating advanced battery technology that ensures long-lasting usage on a single charge. The device would feature a compact battery that provides hours of continuous playback, with a quick charging option for convenience. The neckband would also have energy-saving features to maximize battery life, such as automatic shutoff when not in use or adaptive sound output that adjusts based on environmental noise levels.

In summary, the neckband without earpieces that shoots directed sound combines cutting-edge audio technology, ergonomic design, and safety features to offer a new listening experience. It provides the benefits of personalized, private sound without isolating the wearer from their surroundings. With its combination of comfort, quality,4 and innovation, this device represents a significant advancement in wearable audio technology, meeting the needs of modern consumers who seek a balance between convenience, style, and audio performance.

Claims of the Invention: Neckband Without Earpieces That Shoots Directed Sound

1. **A wearable audio device comprising a neckband designed to sit comfortably around the neck of a user, the neckband comprising:**
   - **Sound projection technology** that generates directed sound beams aimed directly at the ears of the user, eliminating the need for earpieces or earphones.
   - **Miniature speakers or sound emitters** integrated within the neckband structure, configured to emit sound waves in a precise, focused direction toward the user's ears, while minimizing sound leakage into the surrounding environment.
2. **The wearable audio device of claim 1, wherein the sound projection technology utilizes ultrasonic sound waves or beamforming techniques** to focus the audio output in a specific direction, ensuring that only the wearer of the neckband hears the sound clearly without interference from ambient noise.
3. **The wearable audio device of claim 1, wherein the neckband further comprises a flexible, ergonomic design,** ensuring that the device rests comfortably on the user's neck and is suitable for prolonged use, with adjustable components for optimal fit and sound direction.
4. **The wearable audio device of claim 1, wherein the neckband includes wireless communication capabilities** such as Bluetooth, enabling seamless connection to audio sources, including smartphones, tablets, or other compatible devices, for audio playback.

5. **The wearable audio device of claim 1, wherein the sound projection technology is adjustable,** allowing the user to modify the direction and focus of the sound beam based on head position or user preference, enhancing the personalized listening experience.
6. **The wearable audio device of claim 1, wherein the sound emitters are strategically positioned along the neckband** to ensure the sound is directed in a way that minimizes sound leakage and ensures the audio reaches the user's ears with high fidelity and clarity.
7. **The wearable audio device of claim 1, further comprising an integrated power management system,** including a rechargeable battery optimized for extended playback time, with energy-saving features such as automatic shutoff when not in use.
8. **The wearable audio device of claim 1, wherein the neckband is equipped with manual or touch-sensitive controls,** allowing the user to adjust sound volume, audio settings, or sound direction without the need for additional devices.
9. **The wearable audio device of claim 1, wherein the neckband includes sensors** that detect the position or movement of the user's head, automatically adjusting the direction of the sound projection to ensure continuous optimal listening experience as the user moves.
10. **The wearable audio device of claim 1, wherein the device is designed to allow the user to hear the surrounding environment,** ensuring the wearer remains aware of external sounds for safety purposes, particularly in outdoor or public spaces where environmental awareness is critical.
11. **The wearable audio device of claim 1, wherein the sound emitters are capable of reproducing a wide range of frequencies,** providing a high-quality audio experience across the full spectrum, including deep bass and high treble, for an immersive listening experience.
12. **The wearable audio device of claim 1, wherein the device is capable of delivering clear audio output while minimizing distortion,** ensuring an optimal listening experience even in noisy environments or when the user is in motion.
13. **The wearable audio device of claim 1, wherein the neckband further comprises integrated safety features,** such as the ability to pause or adjust the sound playback automatically when the user engages in specific activities (e.g., when speaking or in proximity to loud noises).
14. **The wearable audio device of claim 1, wherein the device is designed to be aesthetically appealing,** with a sleek and modern appearance suitable for everyday wear, combining functionality and style without the bulk or discomfort of traditional headphones or earphones. These claims define the novel aspects of the neckband without earpieces that shoots directed sound, including its sound projection capabilities, ergonomic design, wireless features, and enhanced user control, providing a comprehensive5and personalized audio experience.

Dated this 10th day of March 2022

ABSTRACT

NECKBAND WITHOUT EARPIECES THAT SHOOTS DIRECTED SOUND

[01] The present invention relates to a wearable audio device in the form of a neckband that delivers high-quality, directed sound directly to the user's ears without the need for traditional earphones or earpieces. The neckband incorporates advanced sound projection technology, such as ultrasonic or beamforming techniques, to generate focused sound beams aimed specifically at the listener, ensuring private and clear audio playback while minimizing sound leakage to the surrounding environment.
[02] Designed for comfort, the neckband is lightweight, ergonomic, and adjustable, resting securely around the user's neck for prolonged use. It includes wireless connectivity, such as Bluetooth, for seamless audio streaming from compatible devices. The device further features adjustable sound directionality, volume control, and movement sensors that allow for dynamic adjustments based on the user's head position. The invention ensures that users can enjoy their audio content while maintaining awareness of their surroundings, making it ideal for outdoor activities and public spaces.

This neckband provides a modern, stylish alternative to traditional audio devices, offering an immersive and personalized listening experience while prioritizing comfort and safety.

Dated this 10th day of March 2022